Negative refraction for TM polarization in nanosphere dispersed liquid crystal (NDLC) metamaterial

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ABSTRACT

We discuss the effect of a negative refraction at the interface of uniaxial anisotropic media in the case of nanosphere dispersed liquid crystal (NDLC) matematerial. Finite Element (FE) calculations (COMSOL Multiphysics) are used to trace the propagation of the electromagnetic wave. We show that for chosen values of the parameters of nanospheres and of nematic liquid crystal (NLC) host negative refraction can be obtained for a wide range of incident angles.

1. NANOSPHERE DISPERSED LIQUID CRYSTAL METAMATERIAL

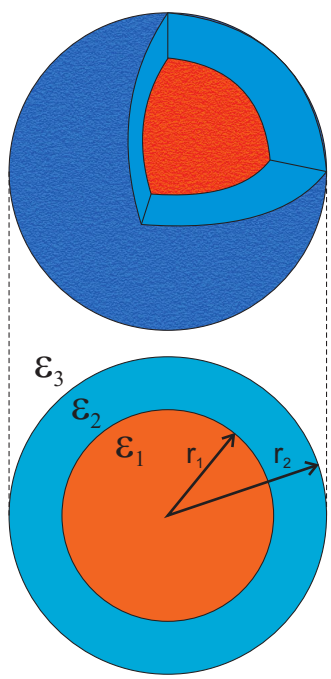

Figure 1. Cross-section of a coated nanosphere.

The concept of the metamaterial used in this paper was put forward by Wheeler *et al.*[1] The material consisting of coated spheres in a host matrix was proposed as a medium with a negative permittivity, permeability and

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refractive index. The possibility of tuning of these electromagnetic properties in such a material using nematic liquid crystal was discussed by Khoo and co–workers[2] and was investigated for chosen values of parameters and over a broad spectral regime.[3–6] In this system the coated spheres were randomly dispersed in NLC host. The control of the effective refractive index of NDLC metamaterial is obtained by means of the reorientation of NLC molecules.

The choice of the parameters is a crucial task for the design of the effective material properties. The resulting effective refractive index of the NDLC media strongly depends on the properties of the two components: core and shell of the coated spheres (Fig. 1) and NLC host.

The effective permittivity and permeability[2] are given by the following expressions:

$$\epsilon_{eff} = \epsilon_3 \left(\frac{k_3^3 + 4\pi i N a_1}{k_3^3 - 2\pi i N a_1} \right), \tag{1}$$

$$\mu_{eff} = \frac{k_3^3 + 4\pi i N b_1}{k_3^3 - 2\pi i N b_1}, \tag{2}$$

where N denotes the number density of the spheres, $k_3 = \sqrt{\epsilon_3} k_0$, where ϵ_3 denotes the permittivity of the liquid crystal host and $k_0 = 2\pi/\lambda_0$. λ_0 denotes the free space wavelength. According to the Mie scattering theory in the dipolar electric and magnetic approximations the amplitudes of scattered electric and magnetic fields are proportional to coefficients a_1 and b_1, respectively; the corresponding expressions can be found in Ref.[7] Those coefficients appear in the expressions for the effective permittivity and permeability of NDLC (1) and (2) and are responsible for the tuning of the resonances. These formulas are valid only for a small value of the number density $N = 3f/4\pi r_2^3$ of nanospheres (f is the filling factor) and in a long–wavelength limit $2\pi/k_3 \gg r_2$.

To ensure the resonance in the effective permeability (2) high values of bulk permittivity of the material of the core have to be used.[1] Relative bulk permittivity of the core with radius r_1 is given by the expression:

$$\epsilon_1(\omega) = \epsilon_1(\infty) \left(1 + \frac{\omega_L^2 - \omega_T^2}{\omega_T^2 - \omega^2 - i\omega\gamma_1} \right), \tag{3}$$

where $\epsilon_1(\infty)$ denotes the high–frequency dielectric constant, γ_1 – the damping coefficient, ω_T and ω_L – transverse and longitudinal phonon frequencies, respectively.

The relative bulk permittivity of the shell with outer radius r_2 is described by the Drude model:

$$\epsilon_2(\omega) = 1 - \frac{\omega_p^2}{\omega^2 + i\omega\gamma_2}, \tag{4}$$

where ω_p denotes plasma frequency and γ_2 – the damping coefficient.

The tunability of the effective permittivity and permeability is provided by changing the permittivity of the NLC host which, for extraordinary polarized wave, depends on the angle θ:[8]

$$\epsilon_3(\theta) = \frac{\epsilon_\parallel \epsilon_\perp}{\epsilon_\parallel \cos^2(\theta) + \epsilon_\perp \sin^2(\theta)}, \tag{5}$$

where $\epsilon_\parallel$ and $\epsilon_\perp$ are the permittivities for the light polarized parallel and perpendicular to the director (or the optical axis), respectively, θ denotes the angle between the directions of the incident light wave vector and the director axis of the NLC.

2. NEGATIVE REFRACTION IN NDLC FOR PLANAR ALIGNMENT OF NLC HOST

We have studied a rectangular NLC cell with the thickness L, filled with NDLC metamaterial (Fig. 2). The parameters of coated nanospheres are given in Table 1. In this paper we analyze a simple case only when the molecules of NLC are parallel to each other and are parallel to the cell boundaries (planar alignment). Ordinary

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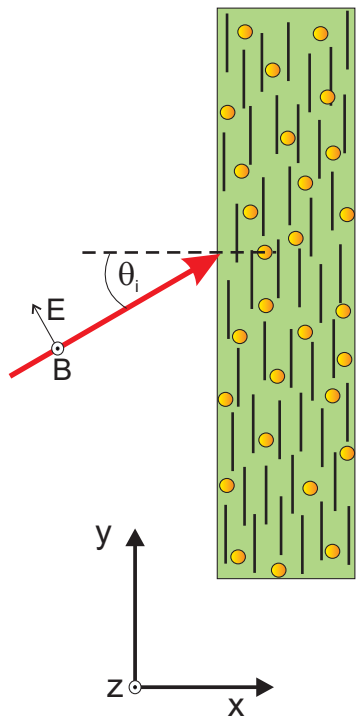

Figure 2. Geometry of the studied system: cell filled with NDLC metamaterial and incident wave.

and extraordinary indices of refraction of NLC molecules were $n_o = 1.41$ and $n_e = 1.7$ which yields $\epsilon_{\parallel} = 2.9$ and $\epsilon_{\perp} = 2$.

The negative refraction was studied using the gaussian wave beam (represented by the Poynting vector) with the wavelength $\lambda_0 = 1.74\,\mu$m. The wave was polarized in $x - y$ plane with magnetic field along z-axis (TM polarization). Effective values of electromagnetic tensor components were calculated using equations from previous section.

r_1	0.05μm
r_2	0.06μm
f	0.06
$\varepsilon(\infty)$	17
$\omega_T/2\pi$	360THz
$\omega_L/2\pi$	855THz
$\gamma_1/2\pi$	3.75THz
$\omega_p\pi$	230THz
γ_2	$\omega_p/60$

Table 1. Parameters used in nanosphere design.

The results of FE calculations of the distribution of magnetic field of incoming and outgoing electromagnetic waves are shown in Fig. 3. We have studied the refraction for a few values of incident angle θ_i: $\theta_i = 20°$ (a), $\theta_i = 30°$ (b), $\theta_i = 40°$ (c), $\theta_i = 50°$ (d), $\theta_i = 60°$ (e). In each case the negative refraction is observed.

3. SUMMARY

To summarize, we have used Finite Element simulations (COMSOL Multiphysics) of the nanosphere dispersed liquid crystal metamaterial to study the hypothetical effect of negative refraction. We have shown that the

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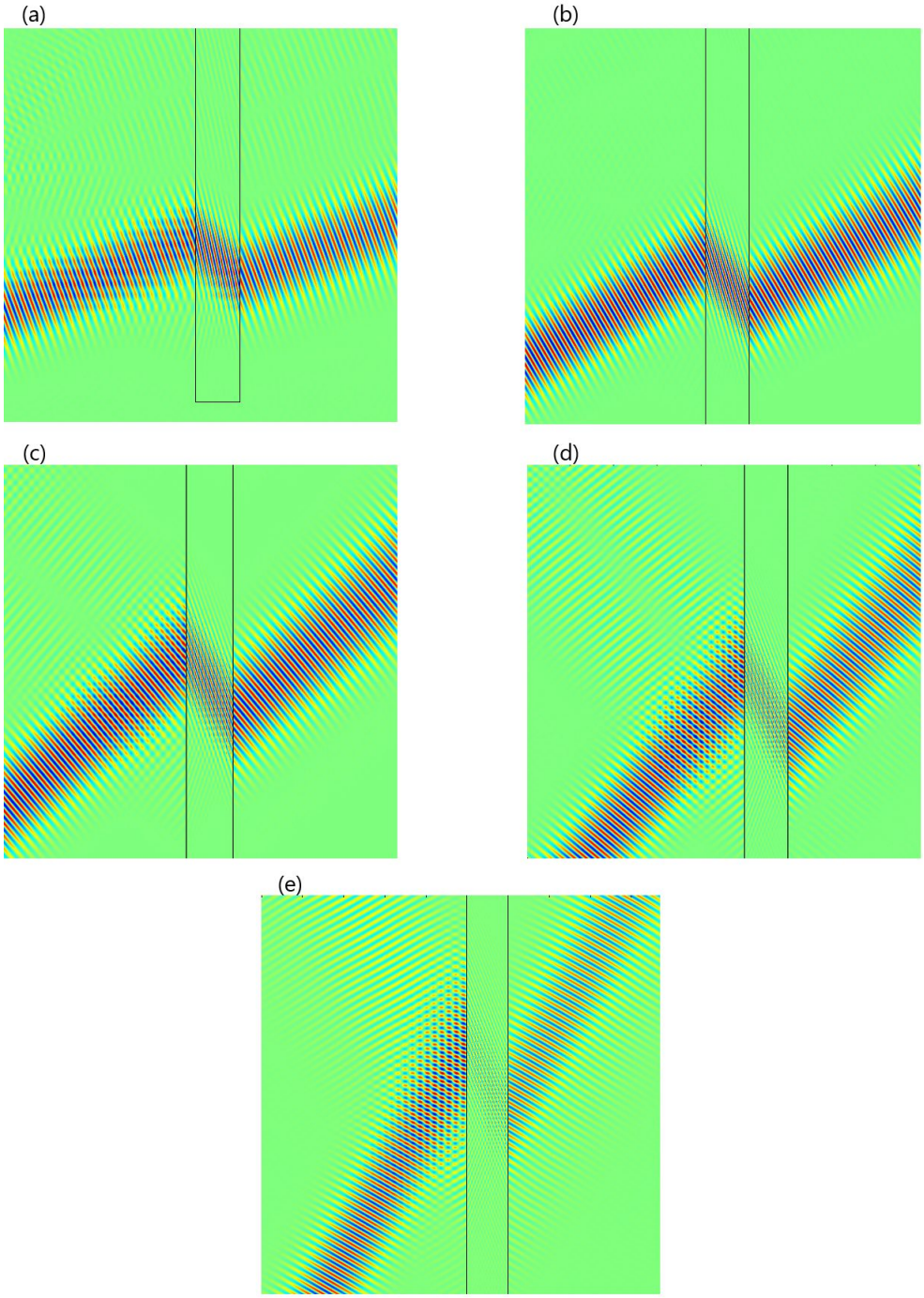

Figure 3. Finite Element calculation of the magnetic field mapping for gaussian beam with TM illumination: $\theta_i = 20°$ (a), $\theta_i = 30°$ (b), $\theta_i = 40°$ (c), $\theta_i = 50°$ (d), $\theta_i = 60°$ (e).

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negative refraction is present for wide range of incident angles. The design can be easily repeated for different sets of parameters and for different configurations of NLC molecules. In particular, the effect of the temperature on negative refraction can be studied using Monte Carlo simulations of the NLC host. Those topics are at progress now and will be published elsewhere.

Acknowledgements

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